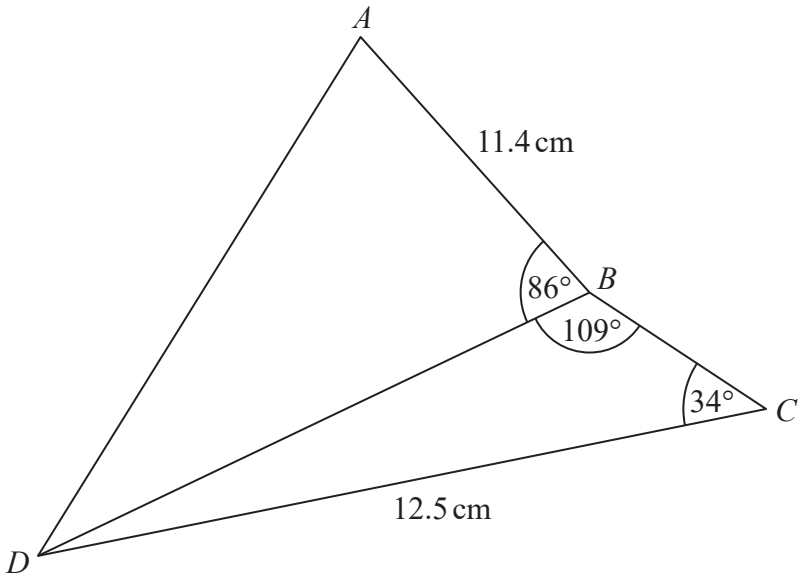
17



Work out the length of AD.
Give your answer correct to 3 significant figures.

First use the sine rule in triangle BCD.

$$\frac{BD}{\sin 34°} = \frac{12.5}{\sin 109°}$$

1 mark
Pair each side with its opposite angle

$BD = 12.5 \times \sin 34° \div \sin 109° = 7.392674744...$ cm

1 mark
Complete the sine-rule calculation

Now use the cosine rule in triangle ABD.

$AD^2 = 11.4^2 + BD^2 - 2 \times 11.4 \times BD \times \cos 86°$

1 mark
Use the two sides and their included angle

$AD^2 = 172.8539781...$; $AD = \sqrt{172.8539781...}$

1 mark
Evaluate in the correct order, then take the square root

$AD = 13.14739434...$ cm. Keep BD unrounded until the end.

1 mark
Round the required length to 3 significant figures

13.1 cm

(Total for Question 17 is 5 marks)